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OPTICAL PHYSICS

Chiral metasurfaces in anisotropic thin film lithium niobate and its nonlinear effect

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Different from conventional two-dimensional photonic metasurfaces, chiral photonic metasurfaces, which are composed of a chiral pattern with subwavelength dimensions, exhibit different chiroptical physical phenomena, such as spin-preserving, nonlinear chiroptical effects, and optical vortex generation. However, the effect of symmetry in the vertical direction on the linear and nonlinear response of the chiral quasi-two-dimensional metasurface has been seldom reported. In this study, we design and simulate a chiral metasurface using topological patterns in lithium niobate (LN) thin film to obtain the optimal circular dichroism and its corresponding nonlinear effect. Triangular holes with a certain rotation constitute a quasi-two-dimensional metasurface, which possesses outstanding chirality for the spin state of light, thus allowing it to control linear and nonlinear chiralities. The nonlinear conversion efficiency η_{SHG} reaches 2×10^{-4} and 4×10^{-3} on the LN structure with substrate and without substrate, respectively. Nonlinear circular dichroism CD_{SHG} is 5.8 dB at 783 nm on a SiO_2 substrate. This tunable chiral metasurface is expected to be applicable to nonlinear chiroptical responses and chiral light modulation. © 2023 Optica Publishing Group

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1. INTRODUCTION

Recently, quasi-two-dimensional (2D) nonlinear dielectric metasurfaces have been proven to possess the ability to control chirally polarized light [1–3]. A quasi-2D structure is composed of patterns carved on nonlinear photonic crystals and exhibits robust circular dichroism (CD) similar to that of the 3D structure [4–6]. In addition, quasi-2D nonlinear chiroptical metasurfaces exhibit optimal CD and chiral nonlinear effects in the infrared range. Therefore, chiroptical metasurfaces that exhibit linear chiral effects and nonlinear characteristics in the infrared range are vital to quantum communication and photonic spin information processing [7–11].

In recent years, the quasi-2D all-dielectric metasurface has received extensive attention. Ultrathin nonlinear crystals exhibit lower loss compared with metal–dielectric structures and nonlinear effects without phase matching [12,13]. However, according to guided-mode resonance (GMR) theory [14], the chiral metasurface depends on the co-excitation of transverse electric (TE) and transverse magnetic (TM) modes in the plane [15]. Anisotropy implies that direction must be considered when designing a pattern and that the same pattern in different directions results in different chiral responses. Since the presence of substrates can also interfere with the co-excitation of guided modes, recent studies by Semnani *et al*. [4] have avoided the effect of vertical symmetry on the chirality by undercutting the quasi-2D chiral metasurface.

The same effect is also shown in the nonlinear process. Nonlinear optical behavior also exhibits chiral characteristics because of the GMR of the metasurface. The most typical phenomenon is chiral second-harmonic generation (SHG). SHG is a typical nonlinear optical behavior [16,17]. Confining light in nonlinear materials is a mainstream method to improve nonlinear response [18–20]. In recent years, as reported for chiral nonlinear materials composed of symmetry-broken metals or crystal patterns, chiral patterns can confine specific polarized light. Chiral metasurfaces are used for practical applications in optical sensing [21–23], polarization conversion, and detection [24–26]. It should be emphasized that asymmetric harmonic generation here refers to asymmetric transmission for left-spin and right-spin illuminations. Such a nonlinear optical behavior is affected by coupling between meta-atoms, which controls the coupling of different polarized light at the fundamental frequency to change the chirality and phase of the SHG wave. For example, a high CD of second-harmonic generation (CD_{SHG})

value was achieved in a plasmonic metasurface, and nonlinear geometric Berry phases were obtained via nth harmonic generation [12,27].

In order to achieve giant chirality and nonlinear responses in thin quasi-2D structure, materials with high refractiveness and a large nonlinear coefficient are required. Lithium niobate (LN) satisfies the conditions above because it possesses outstanding anisotropy and a large second-order nonlinear coefficient ($d_{33} = 19.5$ pm/V at 1313 nm) [28–31]. Hence, it is widely used for optical vortex generation, efficient nonlinear beam shaping, and nonlinear frequency conversion [32,33].

In this study, we design a structure composed of perforating triangular holes in an LN slab, which includes a sandwich structure of LN with air (air–LN–air) and LN with SiO_2 (SiO_2–LN–air). The widely reported platforms on thin film LN (TFLN) have shown its attractive interest since it can provide vertical light confinement in nanoscale. The TFLN platform is more compatible and favorable with the nanofabrication process [34–36]. Meanwhile, TFLN provides a large nonlinear coefficient. To obtain a large CD at the fundamental frequency, we optimize the structural parameter via simulation. We demonstrate that the mode of the electric field is confined in LN, which realizes 4×10^{-3} nonlinear conversion efficiency η_{SHG} on the air–LN–air structure. The electric field excited by different spins is different, and a maximum difference in nonlinear efficiency of 5.8 dB between right- and left-spin light on a SiO_2 substrate is achieved.

2. LINEAR RESPONSES AND BAND STRUCTURE

A schematic illustration of the triangular array is shown in Figs. 1(a) and 1(b). The chirality exhibited by the triangular array to light is demonstrated in the plasmonic structure, whereas light is attenuated during propagation in metallic media. To satisfy the requirement of lossless propagation by infrared light, we designed a periodic topological dielectric structure with an edge length of 930 nm. In order to realize the chirality near 1550 nm, the thickness of LN layer matches the wavelength, and chirality is realized by the patterned structure. Due to the anisotropic property of LN, the size of the pattern can be optimized by analyzing the band diagram. Considering the effect of symmetry, SiO_2 with low refractiveness (ex. air suspended) is selected as the substrate.

For linear CD, the complex transmission coefficients of different polarizations are related to Jones matrix within the circular basis. The chirality of the system is visually reflected by employing Jones matrix. The transmission properties of the chiral metasurface can be described by a 2×2 Jones matrix t, which is shown as follows: Here, $+$ and $-$ in the subscript of the element represent the right and left spin, respectively. The subscript i represents the spin of the incident light, and j represents the spin of the transmission light. The transmission coefficient is equivalent to the transmissivity, $T_{ij} = |t_{ij}|^2$ [37], which can be obtained from the Jones matrix,

$$t = \begin{bmatrix} t_{++} & t_{+-} \\ t_{-+} & t_{--} \end{bmatrix}. \quad (1)$$

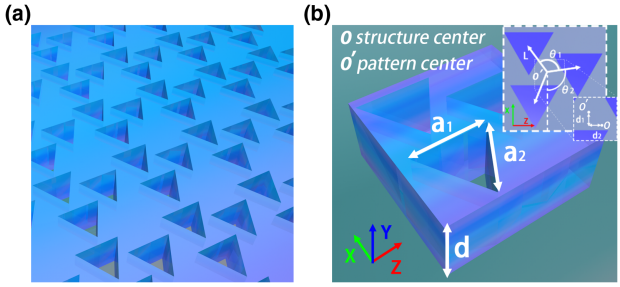


Fig. 1. (a) Schematic diagram of chiral lithium niobate metasurface. (b) The structure consists of triangular holes with chirality, where the structural parameters $a_1 = a_2 = 388$ nm, $\theta_1 = 120°$, $\theta_2 = 96.5°$, $L = 265$ nm, thickness $d = 569$ nm, the deviation between the structure center and the geometric center of the pattern is $d_1 = 70$ nm, $d_2 = 50$ nm, and periodicity 930 nm.

In structural design, the condition $T_{++} + T_{+-} = 1$ (or, for left spin, $T_{--} + T_{-+} = 1$)) must be satisfied; therefore, spin light can pass through and completely reflect the opposite spin light. Notably, the chirality of incident light on different sides of a slab is completely opposite. In the simulation, the side lengths of the triangle, a_1 and a_2, were set as 221 nm, as shown in Fig. 1(b). To simulate the properties of the chiral metasurface, we adopted the finite-difference time- domain method using FDTD (Lumerical Inc.), and the refractive indices of LN were $n_e = 2.13$, $n_o = 2.21$ at 1550 nm [38].

The transmissions of the right and left spin lights are shown in Fig. 2(a), where the blue and red lines represent RCP and LCP lights, respectively. Here, the transmission CD is defined as $\mathrm{CD} = |(T_{++} + T_{+-}) - (T_{-+} + T_{--})|$. The maximum CD is 0.96 at 1551 nm. For LCP, the light achieved a transmission coefficient of approximately 1 at 1551 nm, whereas the transmission coefficient of RCP was approximately 0. To further illustrate the propagation of different circularly polarized lights, the transmission coefficients of the Jones matrix are shown in Figs. 2(c) and 2(d). Figure 2(c) shows that $T_{-+} = 1$ and $T_{--} = 0$ at 1550 nm, which indicates that the chiral metasurface converts left-spin light into right-spin light. The spin reversal characteristics conformed to the behavioral features of all-dielectric planar chiral metasurfaces, thus confirming the chiral transmission process.

The LN thin film requires a hollow substrate for its transport, which imposes high requirements on the structure and the manufacturing process of the underlying material. For practical applications, an appropriate substrate with a low refractive index, such as SiO_2, should be considered. We simulated the transmission of the metasurface repeatedly because the SiO_2 substrate and air layer destroyed the y axis symmetry of the metasurface. However, the frequency of the chiral resonance peak shifted because of the substrate. Therefore, for better chirality, the geometric parameters of the triangular holes were adjusted slightly. As shown in Fig. 2(b), the CD is 0.2 lower than that without the substrate, although a high chirality is maintained (CD = 0.76). This implies that selecting an appropriate substrate does not significantly affect the chiral characteristics of the quasi-2D chiral metasurfaces [14]. In fact, SiO_2 substrate results in incomplete interference between the electric dipole

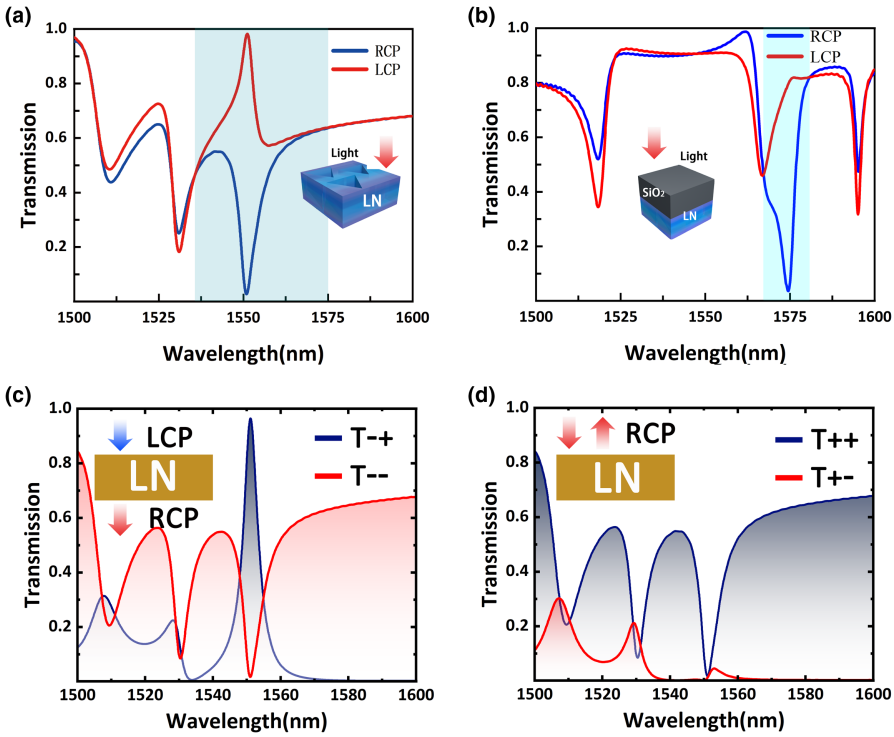


Fig. 2. (a) Power transimission spectra of two spin states of incident light on the metasurface without substrate. (b) Power transimission spectra of two spin states of incident light on the metasurface with SiO_2 substrate. (c) and (d) The transmission coefficients of left spin and right spin light are obtained by simulation, respectively.

and magnetic dipole modes, reducing the chirality of the metasurface. It is difficult to realize perfect asymmetric transmission on the asymmetric structure of the dielectric metasurface, which is caused by incomplete degradation of mode.

The intrinsic chirality of metasurface depends on the interference of electric and magnetic dipoles at resonance frequencies [14]. TE mode and TM mode correspond to the generation of the in-plane electric dipole moment and magnetic dipole moment. The band diagram demonstrates that the judicious design of chiral metasurface results in the degradation of TE and TM modes at the edge of the band diagram. Figure 3(a) shows the band diagrams of the TE and TM modes. The left edge of the band diagram represents the resonance of vertical incident light, where $k = 0$ (denoted by Γ). Low-Q TE and TM modes appear simultaneously in all-dielectric slab structures near the chiral wavelength, and based on the theory of Kerker *et al*. [15,39], a giant asymmetric propagation mode can be achieved by the interference of magnetic and electric dipoles. The metasurface exhibits high TE mode and TM mode resonance at the chiral wavelength (represented by the yellow background), which indicates a stronger electric field at 1550 nm. This shows that the anisotropic LN induces a higher in-plane electric field, which is beneficial for realizing a more effective second-harmonic (SH) process for x-cut LN. To obtain a high SHG efficiency, the electric field can adjust to be confined to the nonlinear material, as shown in Fig. 3(b). Most of the electric field is confined to LN; however, the sharp edges in the triangular structure result in the maximum electric field therein.

3. NONLINEAR CHIRAL RESPONSES

In this section, we discuss the nonlinear response of metasurfaces with chiral properties. SHG was investigated using the parameters mentioned in the previous section (two structures with and without a SiO_2 substrate). The nonlinear coefficient matrix of LN is expressed as follows:

$$d_{ijk} = \begin{bmatrix} 0 & 0 & 0 & 0 & d_{31} & -d_{22} \\ -d_{22} & d_{22} & 0 & d_{31} & 0 & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{bmatrix}. \quad (2)$$

The largest nonlinear coefficient of LN is $d_{33} = -19.5$ pm/V at 1313 nm [12,40], and the nonlinear coefficient on the y axis is $d_{22} = 1.9$ pm/V. The diagonal matrix of the nonlinear coefficients is defined in FDTD Solutions by setting $\chi^{(2)}_{ijk} = 2d_{ijk}$. The nonlinear effect depends on the precise pulse width of the pump light, which is defined as 450 fs, and the light source is a plane wave. The power of the pump light is denoted by P_{FF}, and the power of the SH is denoted by P_{SH}, where $P = \frac{1}{2}\int \mathrm{real}(\vec{S}) \cdot \mathrm{d}\vec{A}$, and $\vec{S}$ is the Poynting vector. The conversion efficiency is defined as $\eta_{\mathrm{SHG}} = \frac{P_{\mathrm{SH}}}{P_{\mathrm{FF}}}$. Because the metasurface is periodic, the power and conversion efficiency of the SHG are calculated based on one periodic structure.

The chiral metasurface with a SiO_2 substrate destroys the y axis symmetry of the structure, which affects the nonlinear response. The nonlinear response of the chiral metasurface with and without the substrate is shown in Figs. 4(a) and 4(b), respectively ($P_{\mathrm{FF}} = 5.309\ \mathrm{GW/cm^2}$). A metasurface with a broken y axis symmetry has a lower P_{SHG} because the high symmetry

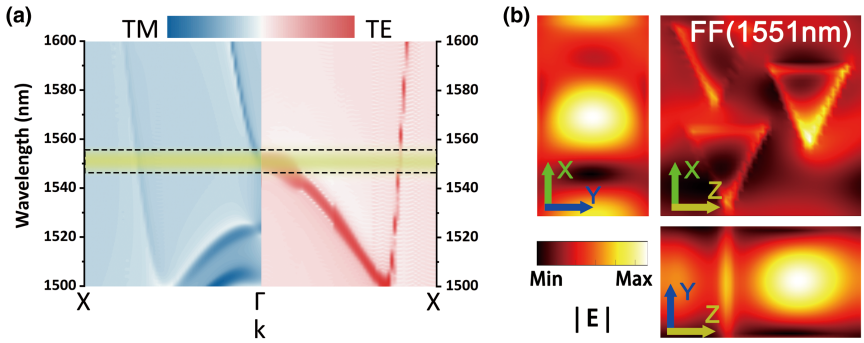


Fig. 3. (a) Band structures of TE and TM modes obtained on air–LN–air structure by FDTD simulation. (b) Electrical field intensity in $x-y$, $x-z$, and $y-z$ planes at the right spin incident wavelength of 1551 nm.

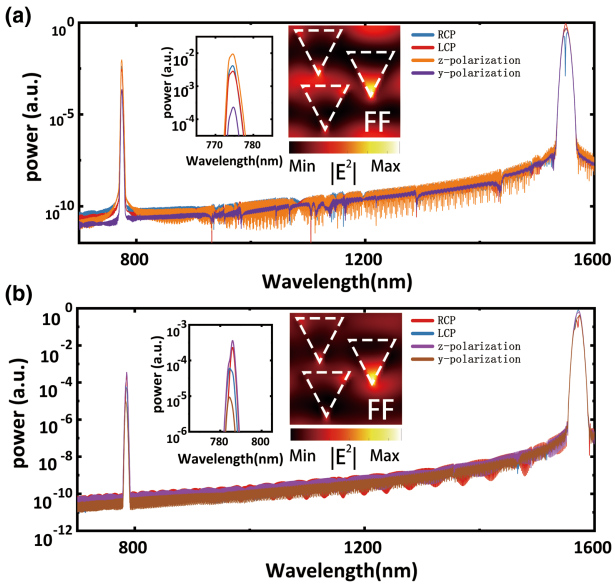


Fig. 4. FDTD nonlinear simulation: normalized nonlinear power transmission spectra of incident light with different polarizations on metasurfaces (a) without substrate and (b) with SiO_2 substrate.

of the light propagation direction is conducive to SHG. The excitation of the resonant mode relies on the polarization of light and the dielectric constant of LN, which is a tensor. In the simulation, the light orthogonal polarization and a phase difference of $\pi/2$ were used as the input, and the amplitude was ensured to be equal to that of the linearly polarized light.

The transmission power spectra of z polarization, x polarization, and left circular polarization (LCP) and right circular polarization (RCP) are shown in the figure, which shows that SHG is affected by incident light of different polarization. The SH power of the x polarization significantly lowers the z polarization. Because the maximum nonlinear coefficient of the y-cut LN film is in the z axis direction. Notably, SHG relies on the chiral enhancement of the resonant mode. Intuitively, $SiO_2-LN-air$ has higher asymmetry, resulting in higher SHG chirality. However, it is important to note that broken symmetry in the propagation direction will cause the degraded modes to split, further weakening the in-plane resonance and reducing the nonlinear conversion efficiency. Higher chirality is at the expense of nonlinear efficiency.

The nonlinear efficiencies of the LCP and RCP with different pump power levels are plotted in Fig. 5(a), which shows that the transmission power of the LCP and RCP SH frequency exhibits chirality. This indicates that, when the pump power increases, the SHG conversion efficiency increases linearly. The SHG contrast is defined as $CD_{SHG} = 10\log|\frac{P_{SHGL}}{P_{SHGR}}|$. Notably, metasurfaces with SiO_2 substrates exhibit higher SHG chirality than air layer metasurfaces, and broken y axis symmetry results in higher chirality for nonlinear processes. The SHG contrast and SHG conversion efficiency chirality can be improved by optimizing the fundamental frequency intensity (highest SHG efficiency $CD_{SHG} = 5.8$ dB). Figure 5(b) shows the CD_{SHG} of the LCP and RCP at different pump light intensities. This indicates that, when the pump power increases, the CD_{SHG} has a slight downward trend.

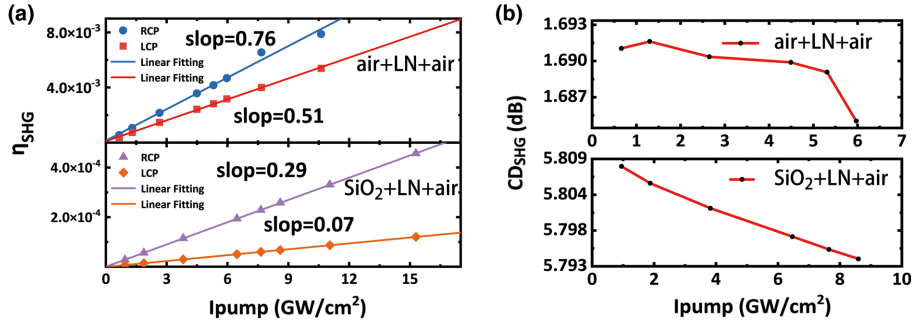


Fig. 5. (a) Nonlinear conversion efficiency under different left spin and right spin incident light intensity on air–LN–air and SiO_2–LN–air, respectively. (b) Circular dichroism power of second-harmonic generation CD_{SHG} on metasurface with SiO_2 substrate and without substrate at 781 nm.

Table 1. Comparison of Chiral Nonlinearity and Nonlinear Conversion Efficiency of Lithium Niobate Materials

Structure	Incidence Mode	Pump Intensity	SHG-CD	η_{SHG}	η_{norm}
LN elliptical cylinders [41]	Oblique	5.3 GW/cm^2	0.98	1.35×10^{-4}	–
Z-shaped LN [42]	Vertical	5.3 GW/cm^2	0.88	1×10^{-5}	–
LN nanodisk [19]	Vertical	5.31 GW/cm^2	–	5.1317×10^{-5}	0.96×10^{-5}
LN nanodisk [43]	Vertical	0.65 GW/cm^2	0.95	1.4×10^{-4}	–
Non-chiral hybrid bilayer [44]	Forward & backward	1.33 GW/cm^2	17.2 dB	0.57×10^{-5}	0.42×10^{-5}
Our structure (air)	Vertical	5.31 GW/cm^2	1.69 dB	4×10^{-3}	7.5×10^{-4}
Our structure (SiO_2)	Vertical	6.47 GW/cm^2	5.8 dB	2×10^{-4}	3.1×10^{-4}

The normalized conversion efficiency η_{norm} is defined as $\eta_{norm} = \frac{P_{SH}}{P_{FF} \times I_{FF}^{peak}} = \eta_{SHG} / I_{FF}^{peak}$, where I_{FF}^{peak} is the fundamental frequency peak intensity. Several nonlinear metasurfaces designed based on LN materials are summarized in Table 1.

4. CONCLUSION

An all-dielectric chiral LN nonlinear metasurface was proposed herein. The all-dielectric structure offers low losses and easy fabrication, and it theoretically realizes a second-order nonlinear process with only intrinsic chirality. The maximum SHG conversion efficiency of the metasurface was approximately 2×10^{-2}%, and CD_{SHG} was 5.8 dB under a pump intensity of 6.67 GW/cm^2. Based on GMR theory, we analyzed the TE and TM band structure distribution of the metasurface at 1551 nm and discovered that the TE and TM band structures degenerate at the chiral wavelength in anisotropic materials. This indicates that a low GMR can similarly generate chirality. Compared with previous studies, this study shows that nonlinear chirality was afforded by only by the intrinsic chirality of the structure and that the effect of chirality on the SHG process was primarily concentrated on the resonance mode, as analyzed based on GMR theory. Nonlinear metasurfaces with simple structures can invert the spin of transmitted light while providing appreciable chirality. Additionally, nanostructures can be flexibly applied to multiple wavelengths by adjusting the pattern parameters and can be applied in nonlinear investigations and device fabrication, such as nonlinear optical modulation, nonlinear quantum optics, and nonlinear sensing.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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